# Mechanized Verification of Hash Table Algorithms: A Comparative Analysis of Quadratic Probing and Open Addressing Paradigms

The formal verification of mutable data structures stands as one of the most rigorously contested and mathematically dense domains within computer science, software engineering, and formal logic. Hash tables, in particular, present a unique confluence of verification challenges due to their absolute reliance on mutable array states, complex memory aliasing, probabilistic time complexity, and highly non-linear memory access patterns. Establishing the total correctness of a hash table requires proving that the underlying array boundaries are never violated, that the hash function distributes elements lawfully, and that the collision resolution strategy guarantees termination without falling into infinite loops. The verification burden is heavily dictated by the specific collision resolution algorithm chosen for the implementation.

While separate chaining isolates collisions into easily verifiable linked lists, open addressing techniques—specifically quadratic probing—require intricate proofs in modular arithmetic and number theory to guarantee that the probing sequence behaves as a complete permutation of the array indices. The mechanization of these proofs requires highly expressive type systems and advanced program logics, such as Concurrent Separation Logic (CSL) or sophisticated SMT-backed verification condition generators. If the probing sequence is not mathematically sound, the algorithm could loop infinitely, which violates the total correctness requirements of interactive theorem provers and automated verification frameworks.

The analysis herein exhaustively documents the verified hash table algorithms across the foremost interactive and automated theorem provers—Rocq (formerly Coq), Lean, Dafny, F*, and Verus—with a specific emphasis on the mathematical mechanization of quadratic probing and precise pointers to the verified source code repositories.[1] Because the explicit directive of this report is to bypass foundational backgrounding of the theorem provers themselves, the exposition will immediately delve into the structural algorithms, the invariant architectures, the termination proofs for open addressing, and the repository locations containing the mechanized source code.

## The Mathematical Topography of Quadratic Probing

To fully comprehend the verification constraints imposed by quadratic probing within formal theorem provers, one must first isolate the mathematical properties that these provers are required to validate. Quadratic probing is an open-addressing collision resolution scheme

wherein the interval between probes increases quadratically, designed specifically to circumvent the primary clustering degradation that plagues linear probing.[6]

Given a hash function $h(k)$ mapping a key $k$ to an initial integer, and a hash table of size $m$, the $i$-th probe position for the key $k$ is rigorously defined by the polynomial function:

$$h(k, i) = (h'(k) + c_1 i + c_2 i^2) \pmod{m}$$

where $c_1$ and $c_2$ are strategically chosen constants, and crucially, $c_2 \neq 0$.[9] If the calculated slot is occupied, the algorithm increments the probe counter $i$ and computes the subsequent index until an empty slot is successfully located or the table is definitively proven to be full.[11]

## Permutation Requirements and Cycle Lengths

The paramount vulnerability of quadratic probing in unverified, imperative environments is the acute risk of an infinite loop. If the probing sequence fails to form a complete mathematical permutation of the table indices, the algorithm may cycle endlessly through a restricted subset of occupied slots, catastrophically failing to locate available empty slots.[10] Formal verification must mathematically eliminate this possibility by proving total termination under all possible state configurations.[13]

When the hash table size $m$ is constrained to a prime number, quadratic probing mathematics guarantees that only the first $\lfloor m/2 \rfloor$ probes in the sequence are completely distinct.[14] If the load factor exceeds $0.5$ (meaning the table is more than half full), there is zero mathematical guarantee that an empty slot will be discovered, mandating immediate dynamic resizing to preserve functionality.[13] Theorem provers verifying prime-sized quadratic probing must therefore encode a strict invariant that the load factor is never permitted to exceed $0.5$ prior to an insertion operation.

However, to guarantee that absolutely every slot in the array is visited (a full cycle), modern high-performance implementations overwhelmingly constrain the table size $m$ to a strict power of two ($m = 2^n$) and utilize the triangular number sequence for calculating offsets.[6] By establishing the constants $c_1 = c_2 = 1/2$, the probing function simplifies to the following optimal form:

$$P(x) = \frac{x^2 + x}{2}$$

This configuration unequivocally ensures that the sequence of offsets $(0, 1, 3, 6, 10, 15, \ldots)$ produces a complete, non-repeating permutation of the indices $[0, 2^n - 1]$.[6]

Mechanizing the proof of this specific property is highly non-trivial and represents a major milestone in data structure verification. As documented in foundational HOL Light proofs underlying the formal analysis of these exact structures, proving that the function $h + P(i) \pmod{2^n}$ is perfectly injective over the bounded domain $[0, 2^n - 1]$ requires proving that if $h + P(i) \equiv h + P(j) \pmod{2^n}$, then it must be true that $i = j$.[16]

The modular equivalence simplifies algebraically to the following Diophantine equation:

$$(i - j)(1 + i + j) = k2^{n+1}$$

Because the terms $(i - j)$ and $(1 + i + j)$ intrinsically possess opposite mathematical parities (one term must invariably be odd while the other is strictly even), and because their absolute magnitudes are strictly bounded by the physical table size, the only valid integer solution within the bounds is $i = j$.[16] Formal verification frameworks, regardless of whether they utilize interactive tactics or SMT automation, must encode this exact number-theoretic proof to successfully satisfy the termination checker for the insert and lookup routines of a quadratic probing hash table.

## Taxonomy of Formally Verified Hash Table Algorithms

| Framework | Algorithm Type | Memory Logic | Primary Repository |
|---|---|---|---|
| Rocq | **OCaml hash table implementation** | Higher-order separation logic (embedded in Coq via CFML / Iris) | Vocal project |
| Lean 4 | **Simple separate-chaining hash table** | Linear usage with bundled well-formedness invariant | Lean Std (mathlib4) |
| *Dafny* | - | - | - |
| F* | **Pulse.Lib.HashTable** | Stratified heap with fractional permissions and resourceful invariants | FStarLang/pulse |
| Verus | **Sharded Hash Table (Ironfleet)** | Verus-verified Rust ownership | verus-lang/verified-ironkv |
| Stainless | **Open addressing with quadratic probing** | Reference swapping construct to prevent aliasing | Scala standard library (LongMap) |

Architectural and algorithmic variations of verified hash tables across major formal verification ecosystems, highlighting the dominant collision resolution strategies.

Data sources: Stainless Paper, Lean 4 Mathlib, Vocal Project Paper, FStarLang/Pulse Repository, Iris Thesis, Verus-lang Repository

# Rocq (Coq): Separation Logic and the VOCaL Project

Within the Rocq (formerly Coq) proof assistant ecosystem, the verification of imperative, heavily stateful algorithms is typically mediated by embedding Separation Logic directly within the Calculus of Inductive Constructions.[18] The most prominent, rigorously exhaustively verified implementation of a hash table in Rocq is found within the VOCaL (Verified OCaml Library) project, which utilizes the CFML (Characteristic Formulae for ML) toolchain developed specifically for extracting and verifying OCaml code.[18]

## Algorithm: Separate Chaining

The formally verified hash table within the VOCaL repository does not utilize quadratic probing; rather, it implements separate chaining.[1] The structural table consists of an array of linked lists (buckets), where each individual bucket stores all key-value pairs that hash to the exact same index.

The deliberate choice of separate chaining over open addressing (such as linear or quadratic probing) in this verified implementation is driven heavily by the mathematical complexities of reasoning about contiguous memory clustering in CFML.[1] In separate chaining, the memory footprint of each bucket is strictly disjoint from the others. Modifying a linked list at index $i$ has zero mathematical probability of altering or corrupting the linked list at index $j$. This spatial disjointness maps perfectly to the separating conjunction operator ($*$) in separation logic.

The Rocq verification proves total functional correctness against a purely functional mathematical map specification. The separation logic invariant explicitly ties the physical heap state of the array and its nested linked lists to the logical, abstract representation of the dictionary. The Coq proof explicitly defines the invariant h ~> TableInState M s, ensuring that the spatial layout in memory correctly and consistently reflects the abstract map M at every stage of execution.[21] The definition heavily utilizes higher-order predicates: Definition h ~> TableInState M s := Hexists d pop init data, h ~> '{ data := d; popu := pop; init := init... }.[21]

To verify the insert operation, the Rocq proof must unfold the TableInState invariant, apply the characteristic formula for the OCaml array update, prove that the hash function maps the key to a valid integer strictly within the bounds of the array, and then fold the invariant back together, ensuring that the new abstract map $M'$ is equivalent to $M \cup \{(k, v)\}$.

## Source Code Pointer

The fully mechanized Rocq source code, including the specific characteristic formulae and the extensive separation logic proofs for the hash table, is located directly in the VOCaL project repository:

| System | Algorithm | Source Code Repository |
|---|---|---|
| **Rocq (Coq)** | Separate Chaining | https://github.com/vocal-project/vocal/tree/master/src/hashtable [18] |

Additionally, the Iris framework (a sophisticated higher-order concurrent separation logic embedded in Coq) contains a verified concurrent hash table implementation. This algorithm similarly leverages separate chaining to handle highly thread-safe operations without necessitating the complex state invalidations and array-shifting mechanisms required by concurrent open addressing schemes.[5]

## Lean 4: Linear Memory, Bundled Invariants, and Lawful Typeclasses

Lean 4 represents a profound paradigm shift in the verification landscape by functioning simultaneously as a highly interactive theorem prover and a high-performance, compiled functional programming language.[24] Because Lean natively enforces pure functional semantics at the language level, mutable state is aggressively managed via uniqueness typing and linear memory access patterns. This allows destructive, in-place updates to arrays (which are essential for hash table performance) without sacrificing referential transparency or mathematical purity.[2]

## Algorithm: Separate Chaining with Power-of-Two Buckets

The standard, fully verified hash table implementation in Lean 4 (located in Std.Data.HashMap or simply Std.HashMap) utilizes a highly optimized separate chaining algorithm.[2] The verification does not currently employ quadratic probing. The foundational data structure consists of a cached size integer and a contiguous array of buckets containing linked lists of key-value pairs.

To ensure optimal computational performance and mathematically sound resizing mechanics, the algorithm enforces a strict invariant that the total number of buckets is unconditionally a power of two. The table automatically doubles its allocated size and rehashes all existing elements when the internal load factor organically exceeds 75% of the bucket count.[2]

The formal verification of this specific structure relies heavily on bundled well-formedness invariants (WF). To perform operations safely without violating memory bounds, Lean requires that the underlying key types mathematically satisfy both the LawfulBEq and LawfulHashable typeclasses.[2]

The LawfulBEq typeclass mandates that the boolean equality operator == forms a true, rigorous equivalence relation (reflexive, symmetric, and transitive), such that $a == b \leftrightarrow a = b$. Crucially, the LawfulHashable typeclass enforces the fundamental hashing property that if $a == b$, then it must strictly follow that $hash(a) == hash(b)$.[2]

Violations of this mathematical property would instantly break the hash table's core lookup invariants, rendering elements unrecoverable.

The Lean 4 implementation meticulously separates the unverified raw array data structure (Std.Data.HashMap.Raw) from the fully verified structure. The verified HashMap wraps the Raw structure alongside a bundled proof of the HashMap.Raw.WF well-formedness invariant. This invariant mathematically guarantees that the cached size perfectly matches the sum of the lengths of all individual bucket lists, and that every single element residing in bucket $i$ legitimately hashes to index $i$ modulo the array capacity.[2]

## Source Code Pointer

The verified Lean 4 hash table implementation, the LawfulHashable typeclasses, and all associated theorems and lemmas are natively available within the Lean Standard Library:

| System | Algorithm | Source Code Repository |
|---|---|---|
| **Lean 4** | Separate Chaining | https://github.com/leanprover/lean4/tree/master/src/Std/Data/HashMap [2] |

While there are numerous unverified, C-based competitive programming snippets floating in external repositories that attempt to link Lean to quadratic probing implementations [26], the mathematically proven algorithm strictly integrated into the Lean standard distribution remains separate chaining.

## Dafny: SMT-Driven Verification of Open Addressing

Dafny is an auto-active verification language heavily utilized in industrial and academic contexts. It compiles directly to executable code (such as C#, Java, or Go) while utilizing the Z3 SMT solver to automatically discharge highly complex proof obligations based on user-provided preconditions (requires), postconditions (ensures), and detailed loop invariants (invariant).[27]

## Algorithm: Open Addressing with Linear Probing

While strict quadratic probing is exceptionally rare in standard libraries due to SMT non-linear arithmetic limitations, the full verification of open-addressed hash tables has been successfully and rigorously achieved in Dafny. A landmark case study by Hance et al. (and further expanded by De Boer et al.) involved the formal verification of a complex system structurally akin to the

JDK's IdentityHashTable.[1] This algorithm utilizes open addressing with linear probing.

In linear probing, upon detecting a collision, the algorithm steps through the array sequentially ( $index + 1, index + 2, \ldots$ ) until an empty bucket is found. The Dafny proof requires incredibly explicit loop invariants to mathematically demonstrate that the probing sequence correctly wraps around the array boundaries using modulo arithmetic, and that the internal while loop will absolutely terminate.

Termination in Dafny is proven by providing a decreases clause. For open addressing, this requires maintaining a strict, unyielding upper bound on the load factor, ensuring the table is never completely full. The prover must be mathematically convinced that an empty slot (or a slot containing a matching key) will invariably be encountered within a bounded number of steps strictly equal to the capacity of the table.[17] If the table were allowed to reach 100% capacity, the linear probe could loop infinitely, and Dafny's SMT solver would immediately reject the insert function for failing termination checks.

The verification effort in Dafny elegantly isolates functional correctness from heap manipulation. The engineers first defined the hash table conceptually using immutable, mathematically pure data structures (such as Dafny's native map or seq), proving the mathematical model in isolation. Subsequently, they mapped the highly mutable imperative array implementation to this verified mathematical model using Dafny's ghost variables and specification annotations.[1]

## Mechanizing Quadratic Probing Models

Although the standard verified structures use linear probing, the exact permutation mathematics required for quadratic probing have been modeled and verified in Dafny for educational and advanced research purposes.[16] Specifically, the proofs confirming that the sequence $(x^2 + x)/2 \pmod{2^n}$ does not miss any free slots have been successfully discharged by Dafny's verifier. By providing the SMT solver with the necessary inductive lemmas regarding the parity of $(i - j)$ and $(1 + i + j)$, Dafny can successfully verify that quadratic probing termination is guaranteed when the table size is a power of two.[16]

## Source Code Pointer

The source code for the Dafny-verified linear probing hash table models, as well as the diverse benchmarking of collision resolution algorithms, can be located within the Dafny core repositories and the expansive DafnyBench dataset:

| System | Algorithm | Source Code Repository |
|---|---|---|
| **Dafny** | Linear Probing (Open Addressing) | https://github.com/dafny-lang/dafny and DafnyBench https://namin.seas.harvard.edu/pubs/dafnybench.pdf [29] |

# F* (FStar): Pulse and Stateful Memory Verification

F* is a sophisticated dependently typed programming language and interactive proof assistant specifically designed for verifying the strict functional correctness and deep security properties of low-level systems code. Recent, groundbreaking advancements in the F* ecosystem have introduced Pulse, a dedicated separation logic framework meticulously tailored for verifying concurrent and highly stateful programs directly within the F* environment without relying entirely on Z3 refinement types.[4]

# Algorithm: Stateful Imperative Hash Tables

The Pulse library within F* features a dedicated, fully verified Pulse.Lib.HashTable module. The specific collision resolution strategy implemented in the core Pulse library is a stateful, imperative hash table, which is primarily optimized for proving strict memory safety, spatial disjointness, and absolute data race freedom in low-level memory environments, rather than focusing purely on complex probing sequences like quadratic offset generation.[4]

The mathematical proofs inside Pulse rely heavily on fractional permissions and localized points-to (pts_to) assertions. These assertions explicitly guarantee that the raw array indices manipulated during the hashing and insertion phases strictly conform to the memory bounds of the allocation, entirely preventing buffer overflows.[4] Because F* is often used to verify cryptographic primitives and secure network stacks (such as Project Everest), the hash table verification prioritizes constant-time access patterns and deterministic memory safety over maximizing load factors via complex open addressing sequences.

In Pulse, the physical state of the hash table array is connected to a logical, erased FStar.Map representation. The insert and lookup operations are verified by demonstrating that any mutation to the pts_to array slice mathematically corresponds to an identical update in the logical FStar.Map.

# Source Code Pointer

The verified source code for the F* Pulse stateful hash table, complete with its separation logic invariants and pts_to proofs, is located directly in the Pulse library repository:

| System | Algorithm | Source Code Repository |
|---|---|---|
| *F (FStar)** | Stateful Hash Table (Pulse) | https://github.com/FStarLang/pulse/tree/main/lib/pulse/lib/Pulse.Lib.HashTable [4] |

# Verus: Rust Borrow Semantics and Sharded Architectures

Verus is a state-of-the-art formal verification tool designed specifically for the Rust programming language. It represents a massive leap in verification ergonomics by natively leveraging Rust's strict ownership, lifetime, and borrowing semantics, automatically translating them into verification conditions that are discharged by the Z3 SMT solver.[30] Because the Rust compiler itself already guarantees the absolute absence of data races and memory aliasing, Verus is liberated to focus almost entirely on deep functional correctness and complex structural invariants.[30]

# Algorithm: Sharded Hash Table (IronSHT)

A prominent, rigorously verified hash table implementation in Verus is found in the verified-ironkv project, which implements the IronFleet Sharded Hash Table (IronSHT) key-value store.[32] To robustly support highly concurrent distributed workloads, the algorithm inherently relies on a sharded architecture rather than a single monolithic open-addressed array.

Verus elegantly handles the formal verification of mutable operations (such as acquiring a borrow_mut on a specific hash table bucket) by utilizing complex prophecy variables or specific encoding pairs (old(v), new(v)).[3] When an element is inserted or heavily mutated, Verus mathematically proves that the abstract, logical view of the entire hash table transitions cleanly to a new mathematical set that includes the mutated element, while flawlessly preserving the integrity of all other non-aliased elements residing in the table.[3]

For example, the postcondition for a mutable borrow function in Verus is explicitly defined to ensure the logical view matches the physical state: ensures self.view() == old(self).view().set(key, *ref.value()).[3]

# The Quadratic Probing Reality of SwissTable

While the Verus-verified IronSHT utilizes sharding, it is critically important to note the underlying reality of the Rust standard library std::collections::HashMap, which Verus targets for base functionality. The Rust standard library explicitly utilizes the SwissTable architecture.[34] SwissTable is a highly sophisticated open-addressing scheme that natively utilizes quadratic probing combined with 16-way SIMD (Single Instruction, Multiple Data) metadata lookups to

achieve unprecedented cache locality and performance.[34]

The quadratic probing mechanism in Rust's SwissTable is designed to provide extreme resistance against HashDoS attacks by utilizing randomly seeded SipHash 1-3 algorithms in conjunction with the quadratic offset.[34] While the total, line-by-line formal verification of the raw SIMD-optimized SwissTable implementation remains an active, ongoing frontier in the Verus ecosystem, the higher-level structural invariants of the standard bucket-based and sharded tables have been successfully proven.[3]

## Source Code Pointer

The Verus-verified IronSHT implementation and the specific verification logic for mutable container borrows are located at:

| System | Algorithm | Source Code Repository |
|---|---|---|
| **Verus** | Sharded Hash Table (IronSHT) | https://github.com/verus-lang/verified-ironkv [32] |

## The Stainless Exception: Formally Verified Quadratic Probing

While Rocq, Lean, Dafny, and Verus predominantly feature verified implementations of separate chaining, linear probing, or sharded models due to their mathematically simpler verification conditions, the specific, exact requirement of verifying an imperative **quadratic probing** array algorithm has been definitively and beautifully solved in the Scala ecosystem using the Stainless program verifier.[1]

Because the original query explicitly demands verified algorithms for hash tables with a precise focus on *quadratic probing*, the Stainless verification of Scala's LongMap must be highlighted as the pinnacle achievement in this specific algorithmic niche.

## Algorithm: Open Addressing with Strict Quadratic Probing (LongMap)

A monumental verification case study conducted by Chassot and Kuncak involved the complete, uncompromising formal verification of the highly mutable LongMap from the standard Scala library.[1] LongMap is a high-performance, primitive-keyed hash table that utilizes open addressing strictly within a single contiguous array and exclusively employs quadratic

probing to resolve all internal collisions.[1]

The formal verification of this specific structure in Stainless directly addresses the extreme mathematical complexities outlined in the opening sections of this report. The algorithm fundamentally handles collisions using a recursive quadratic function specifically designed for the probe sequence:

$$index_{x+1} = (index_x + 2 \times (x+1) \times x - 3) \ \& \ mask$$

where the variable mask is rigidly defined as $N - 1$, structurally enforcing at the type level that the table size $N$ is unconditionally a power of two ($N = 2^n$) at all times.[1]

To effectively mechanize this inside the Stainless verifier, the framework requires proving beyond any mathematical doubt that the internal seekEntry and seekEntryOrOpen routing algorithms will inevitably terminate. The proof systematically establishes that the bitwise AND operation (& mask) perfectly replicates modulo $2^n$ arithmetic without overflow, and that the specific quadratic polynomial chosen forms a complete, unbroken cyclic permutation over the physical array.[1]

## The Complexity of Tombstone Deletion

Furthermore, the Stainless verification successfully manages the notorious complexities of "tombstone" deletion inside quadratic probing sequences. In open addressing, simply clearing a slot to delete an item fundamentally breaks the probe sequence for all subsequent elements that collided and were shifted further down the quadratic chain.[37]

The formally verified LongMap successfully bypasses this by utilizing a designated tombstone value (Long.MinValue) to permanently mark deleted slots, ensuring that the quadratic probe chain remains mathematically unbroken during subsequent lookup operations.[1] The data structure representation invariant (valid) is exhaustively proven to hold across all highly mutable operations (insert, delete, repack), mathematically guaranteeing that absolutely no elements are ever lost to secondary clustering or broken probing chains.[1]

## Source Code Pointer

The formal specification and fully verified source code for the quadratic probing LongMap algorithm are meticulously maintained by the EPFL LARA group within the core Stainless verification repository:

| System | Algorithm | Source Code Repository |
|---|---|---|
| **Stainless (Scala)** | Quadratic Probing | https://github.com/epfl-lara/stainless [1] |

# Algorithmic Verification Trade-Offs and Implications

The expansive landscape of mechanized hash table verification reveals a stark, fascinating dichotomy between the algorithms preferred by systems software engineers and those preferred by formal verification engineers.

In unverified, high-performance computing environments (such as standard library development for C++, Rust, and Go), open addressing frameworks utilizing quadratic probing, double hashing, or SIMD-accelerated metadata architectures are overwhelmingly preferred. These algorithms inherently provide superior CPU cache locality, complete elimination of pointer-chasing overhead, and strict contiguous memory layouts.[35]

Conversely, within the rigid realm of formal verification using systems like Rocq, Lean 4, Dafny, and F*, the predominant, overwhelmingly favored paradigm remains separate chaining. The underlying verification conditions required to prove the correctness of separate chaining are fundamentally, mathematically simpler. The SMT solver or interactive tactic script only needs to reason about the isolated memory bounds of a single host array and the inductive, recursive properties of strictly disjoint linked lists.[2] The mathematical burden of proving that a specific quadratic polynomial yields a perfect bijective mapping over an integer ring modulo $2^n$ places significant, often insurmountable strain on automated SMT solvers and requires extensive, highly manual tactic scripting in interactive provers.

The successful, comprehensive verification of the quadratic-probing LongMap via the Stainless verifier [1] stands as a definitive, landmark proof-of-concept that complex, cache-optimized open addressing schemes can indeed be formally mechanized without sacrificing performance. As modern verification tools like Verus continue to exponentially mature—by leveraging Rust's native borrow checker to automatically discharge complex aliasing conditions before they ever reach the SMT solver—the historical barrier to verifying heavily optimized quadratic and SIMD-based probing algorithms will continue to lower. This trajectory promises to permanently bridge the longstanding gap between theoretical verification constraints and production-grade hardware efficiency, ultimately allowing quadratic probing to become the standard, formally verified choice across all modern theorem proving ecosystems.

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